Cubic five-gluing: exact source census and proof checks

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The Rust extension identifies **252 exact binary source models**, grouped into **126 quadratic-twist pairs**, that pass the five-torsion criterion and fail the earlier three-torsion criterion over F512/F8. All 126 pairs also pass the small-geometric-isogeny exclusions. The written proof gives a genus-three Jacobian isogeny for at least one source twist in each pair. **A novel affected family and an explicit log transfer are not established.**

This continues the user's search for a novel criterion and proof. Beating rho is not a gate. The experiment code is Rust in cryptanalysis; Python is used only for command receipts and document/figure rendering.

Exact results

Check	Observed result	Scope
Weighted Lagrangians at ell=5	19,656 per form	all four nonzero q residues
Transverse Lagrangians	17,280 per form	coordinate intersections zero
GL2(F5) matrices checked	480	all compatible q, using det M=q cubed
Positive matrices	184	180 split-distinct, four scalar
Criterion/enumeration disagreements	0	independent RREF subspace enumeration
Positive ell=11 constructions	5,950	constructive checks; negative cases not enumerated
Full-absolute-degree source models	1,008	a=0,1 and all 504 full-degree b in F512
Five-positive, three-negative source models	252	complete exact model census
Small-isogeny screen	126/126 twist pairs pass	degree 1 through 6 excluded geometrically
Rust release tests	39 pass	all targets, including existing sweeper tests
Lean fixed certificate	five declarations pass	general classification and geometry unformalized
Transitive Lean audit	eight theorems, zero failures	includes three generated structure theorems; no native trust

The 1,008 sources divide equally into four groups: both torsion screens positive, only three positive, only five positive, and both negative: 252 models in each. These are exact finite counts, not sampled yields. The five-only models have traces $\pm 7, \pm 13, \pm 17, \pm 23, \pm 37, \pm 43$. They represent 14 absolute Frobenius orbits of coefficients, not 252 unrelated families. All field arithmetic comparisons agree between the log-table and raw polynomial paths. Screening/counting logic is shared; there is no external independent implementation review.

One concrete twist pair

Use $K = \mathbb{F}_2[z]/(z^9 + z + 1)$, encoded by modulus 515, and $E_a : y^2 + xy = x^3 + ax^2 + 48$ with a=0 or 1. In this basis $j = 350$, $j^8 = 70$.

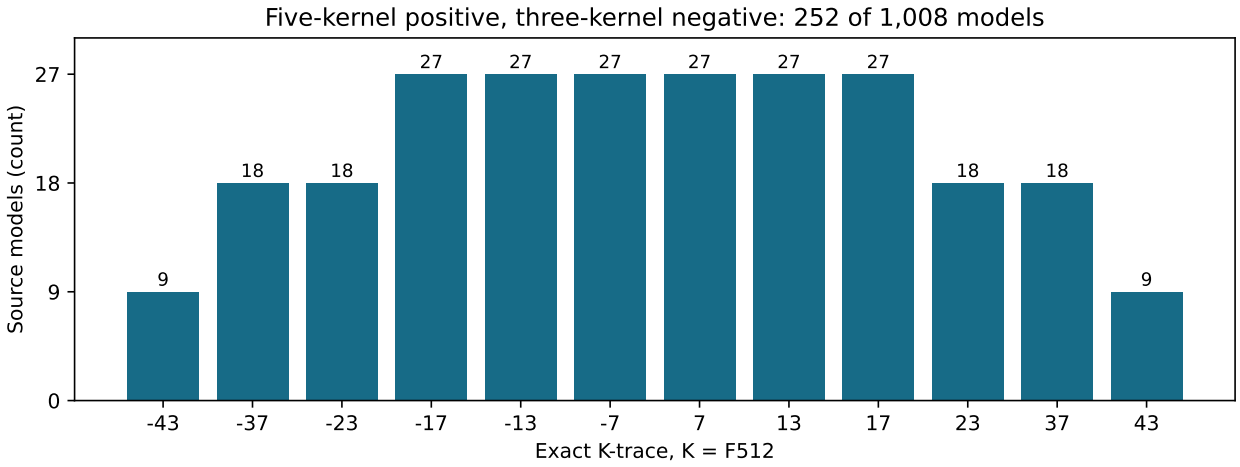


Figure 1: Exact counts by source trace. All displayed models pass the geometric screen; the existence theorem selects one unknown twist per pair. No performance or large-field yield is inferred.

Model	Exact source order	Trace	Prime factors
a=0, b=48	520	-7	2 cubed times 5 times 13
a=1, b=48	506	7	2 times 11 times 23

The modular values, in order $\Phi_3(j, j^8), \Phi_5(j, j^8), \Phi_3(j^2, j^8), \Phi_3(j, (j^8)^2)$, are **24, 306, 398, 201**. All are nonzero, and the Frobenius-distance tests exclude degrees 1,2,4. Thus the sufficient geometric criterion holds. It gives existence for one twist, preserving its prime subgroups away from five. It does not select that twist or supply target coordinates. No generator was selected, so the EC1 subgroup identity remains null.

The illustrative canonical module matrix $M = \mathrm{diag}(3, 4)$, $q = 3$ modulo five, has basis vectors $(1, 0, 3, 0, 4, 0)$, $(0, 4, 0, 4, 0, 0)$, $(0, 1, 0, 0, 0, 4)$. The Lean certificate checks independence, Frobenius stability, pairwise basis isotropy and coordinate transversality for exactly this module. It does not instantiate these vectors as points on $E[5]$.

Proof and novelty status

The proof chain is the module theorem, then the indecomposability lemma and binary criterion. Both are derived written proofs. The fixed finite module is checked in Lean 4.34.1, commit 5045d0056413266e57c625dcd7c365b10e377c52, with no external packages. Each of the five named declarations depends only on propext. The transitive audit also covers the generated theorems. The proof map states the missing general and geometric bridges.

Tian's isotropic-quotient approach is published prior work; Hess's generalized descent already gives broad existence for these binary sources. The inspected Tian 2020 preprint assumes a suitable kernel rather than stating this exact classification. This is not a novelty certification: the full 2023 journal article remains unread, and absence in web searches does not establish priority. The 2018 cofactor-two abstract describes an odd-characteristic construction. These sources do not justify a claim that our exact binary criterion is new. No weak-curve catalogue entry or verified isogeny-route edge has been added.

Reproduction and retained evidence

From the repository root, with the pinned Rust environment:

```
cargo run --release --locked --manifest-path experiments/ghs-sweeper/Cargo.toml \
  --example cubic_gluing -- exhaustive
cargo run --release --locked --manifest-path experiments/ghs-sweeper/Cargo.toml \
  --example cubic_gluing -- binary-census
```

```
cargo run --release --locked --manifest-path experiments/ghs-sweeper/Cargo.toml \
  --example five_gluing_geometry
```

For Lean, run `lake build` in `formal/`. The repository corpus gate also builds and audits it alongside all previously covered sources.

The module/census contract and Rust source were frozen at `0c803777`; the partial Lean project at `78bda02c`; the geometric check and proof at `b71fec1c`. `cubic-gluing-checks/census-commands.json` and `geometry-commands.json` retain exact argument arrays, source and output digests, exits, revisions and execution durations. Durations are receipts, not comparable performance measurements. Large JSON output is stored as lossless deterministic gzip, with both stored and uncompressed hashes.

Formatting and Clippy with warnings denied passed. All 39 release tests passed after adding the geometry checks. The old 32-file Hess, 30-file three-gluing and 63-file kernel manifests remain separate frozen evidence. The core sweeper, dependencies and checkpoint fingerprints are unchanged. The 355 older denominator jobs were not resumed by these separate checks. The artifact policy explicitly retains the three pre-existing frozen report PDFs (78–94 kB each), preserving their manifest bytes during integration onto main. New reports and compressed outputs fit the normal size thresholds.

The first full Lean-corpus replay failed because this sparse checkout lacked the tracked `src/curve.c` needed by an existing GLV source-binding check. A second replay exposed a stale pre-existing live-source pin: main's later C changes altered the whole-file hash while preserving the exact table selector and guards. Both failures are retained. The new gate preserves the old receipt, replays its historical source and both negative controls, then checks the current main source separately with the same selector assertions and negative controls. No C code or historical GLV artifact was changed. The third replay passes for all 95 research Lean sources; the new dated corpus receipt binds the current and historical source hashes separately.

The older scope, kernel and criterion figures were checked and retain their original scope. New diagrams show the five-gluing hypotheses and unfinished reconstruction. Canonical curve/isogeny graphs remain unchanged.

Remaining work against the request

Requirement	Status
New recognition criterion with written proof	Derived and exactly checked in the stated finite cases; publication priority unresolved
Rust sweeper extension in crypto/cryptanalysis	Implemented and locally verified
Novel affected curve family	Not established
Full Lean proof	Partial fixed case only
Explicit selected genus-three target and subgroup/log transfer	Not constructed or performed
Construction, transfer and solver costs	Not measured; not replaced by the short check runtimes

The next construction target is the b=48 twist pair. It requires actual five-torsion coordinates, a characteristic-two quotient reconstruction, arithmetic twist selection and an evaluated prime-subgroup isogeny. The specific exit conditions remain in the problem record.